

Lecture 1: Real Numbers

AMA3707 Real Analysis

1 Numbers

Definition 1.1 (Natural Numbers). The *natural numbers* are the positive integers:

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

In these notes, $0 \notin \mathbb{N}$.

Definition 1.2 (Integers). The *integers* are the whole numbers, including the positive integers, the negative integers, and zero:

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}.$$

Definition 1.3 (Rational Numbers). A real number is called *rational* if it can be expressed as a quotient of two integers. The set of rational numbers is

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

Remark 1.1. Under the usual identifications, $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q}$.

2 Fields

Definition 2.1 (Closed Operation). An operation $*$ is said to be *closed* on a set A if for all $a, b \in A$, we have $a * b \in A$.

Definition 2.2 (Field). A *field* is a set of numbers or elements where addition and multiplication are closed operations.